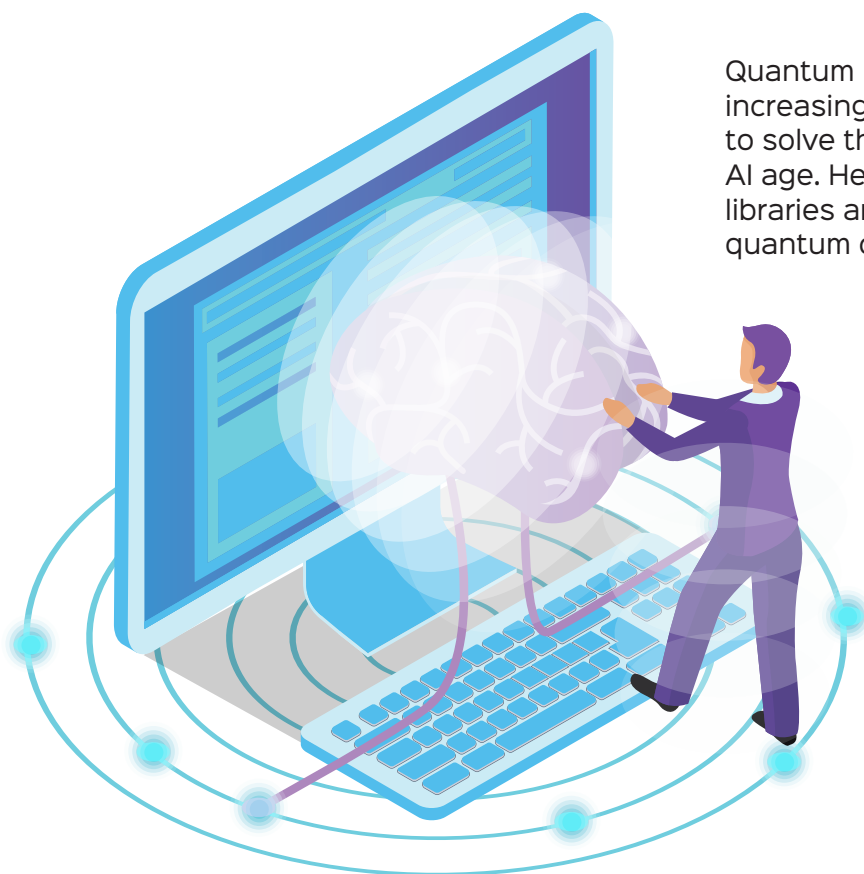


Popular Open Source Toolkits for Quantum Machine Learning

Quantum machine learning is becoming increasingly popular due to its ability to solve the complex problems of the AI age. Here are a few open source libraries and frameworks that help with quantum computations.



of 20.1% from 2024 to 2030. In another report by *The Quantum Insider*, the worldwide quantum sector could offer more than US\$ 1 trillion to the world economy between 2025 and 2035. During this period, corporations in the quantum space are expected to get US\$ 50 billion in income. This growth in quantum computing will enable about 250,000 new jobs by 2030, moving to 840,000 by 2035.

While this field is still in its development stages, the expected financial ramifications are critical. Specialists anticipate that as quantum computing develops, it will open new pathways for ventures, especially in regions requiring high-accuracy computations.

Quantum machine learning (QML)

Quantum machine learning (QML) is the key technology consolidating the standards of quantum mechanics with AI algorithms for information processing. The non-traditional properties of quantum computations like superposition, entanglement and quantum interference give huge computational edges, especially in

Quantum computing is a key area of research and development nowadays in corporate as well as government projects. Traditional computations depend on bits that encode data in parallel conditions of 0s and 1s, while quantum systems use qubits that can exist in numerous states thanks to the features of superposition and entanglement. This ability empowers quantum computers to handle estimations of extraordinary intricacy, making them particularly appropriate

for high performance applications like cryptography, drug discovery, complex numerical computations, healthcare, and more.

The worldwide quantum processing market is supposed to develop significantly over the next 10 years. IBM, Google, and D-Wave are the big companies working on various deployments and algorithms related to quantum models and applications. According to *Grand View Research*, the worldwide quantum computing market is projected to grow at a CAGR